



Planning estimates for discussion with qualified parties. Not guarantees of future performance. Not an offer to sell securities.

1. Executive summary

TITAN-X is a free-flight stratospheric aerostat with a 20,000 lb modular payload, indefinite endurance on helium buoyancy and solar power, and mission reconfiguration by swapping gondolas and underbelly modules.

Investment thesis: fund pathfinder → first full-scale hull → California firefighting + metro nodes → planning 12-unit fleet by 2030. Pre-positioning converts response from days to hours; civil and commercial utilization underwrites year-round availability; dual-use configs share one cost base.

\$13M Unit CAPEX (fleet volume)	\$156M 12-unit fleet CAPEX	\$2.4M Annual OPEX / unit	\$200K Monthly break-even / unit
17.8 : 1 Planning 10-yr fleet ROI	20,000 lb Net modular payload		

2. Platform at a glance

PARAMETER	VALUE	NOTES
Size	480 ft × 200 ft	Swept manta planform
Net payload	20,000 lbs	≈ school-bus weight
Altitude range	500 – 130,000 ft	Tree-top to lower stratosphere
Endurance	Indefinite	Solar + buoyancy
Ops carbon (persistent)	Near-zero	No combustion for loiter
Architecture	Fully modular	Gondola / underbelly swap

- Rockoon loft to ~100,000 ft for propellant savings, then recover the platform
- Civil, commercial, and defense requirements share the same platform and cost base



Firefighting configuration — retardant and water delivery with sensors and incident-command packages. California is the planning early concentration.

3. Problem & solution

Wildfire and disaster response still depends on scarce surge aircraft. Satellites persist but cannot deliver mass. Cities need hours-scale airborne nodes for observation, temporary communications, and logistics.

TITAN-X fills the band between aircraft and satellites: indefinite loiter, school-bus-class modular payload, near-zero persistent ops carbon, and pre-positioned regional coverage instead of surge-only response.

4. Regional deployment

Major cities & high-value regions

One TITAN-X (or a pair) per region, loaded with modular response packages, on station in hours — disaster response, communications restoration, observation, logistics.

California firefighting concentration

- Fire-behavior, thermal, and smoke-plume sensors
- Retardant or water delivery modules
- Incident command and communications packages
- Decision-support for incident commanders

Modules shift to other missions in lower-risk periods to preserve utilization.

Smaller complementary platforms

Secondary observation, relay, weather, and lower-cost coverage under full-size regional nodes.

5. Customers & missions

SEGMENT	USE
State / regional wildfire authorities	CA firefighting concentration
Metro & high-value regions	Persistent disaster & comms nodes
Critical infrastructure	Monitoring and logistics support
Maritime / shipping	Domain awareness and ship support
Public safety	Airborne threat awareness (low-cost drones)
Science / weather	High-altitude observation
Rockoon / launch customers	Stratospheric loft + recover
Dual-use / defense (selected)	ISR, relay, forward logistics, responsive access

6. Unit economics

ITEM	PLANNING VALUE
Unit CAPEX (fleet volume)	\$13.0M
12-unit fleet CAPEX	\$156M
Annual OPEX per unit	\$2.4M
Break-even utilization	~\$200K / month
Mature utilization target (base)	\$8–12M / unit / year
Contribution at \$10M rev / \$2.4M opex	~\$7.6M / unit / year

Illustrative mature revenue mix

STREAM	SHARE
Regional standby / availability	35–45%
Wildfire / disaster mission fees	20–30%
Infrastructure / maritime / public safety	15–25%
Rockoon / science / specialty	5–15%
Dual-use / defense options	0–20%

7. Financial plan 2026–2030

USD millions. Anchored to consolidated summary CAPEX / OPEX / break-even. The planning 17.8 : 1 ten-year fleet ROI is a long-horizon target on the stood-up fleet — not a 2030 cash guarantee.

Fleet build (base)

	2026	2027	2028	2029	2030
Full-size units EOY	0*	1	3	6	12
CA firefighting-primary	—	1	2	3	4
Metro / regional nodes	—	0	1	3	8

*Pathfinder year — endurance, station-keeping, modular swap validation.

Capital expenditure (base, \$M)

	2026	2027	2028	2029	2030	TOTAL
Pathfinder + engineering	18	6	4	3	2	33
Hull production (learning curve)	—	18	30	42	78	168
Ground / ops infrastructure	4	6	8	10	12	40
Total CAPEX	22	30	42	55	92	241

Operating model (base, \$M)

	2026	2027	2028	2029	2030
Avg units in service	0	0.5	2.0	4.5	9.0
Revenue	0.5	4	18	42	90
Direct OPEX	1.5	3.5	7	13	24
Program / G&A	4	5	7	9	11
EBITDA (planning)	(5.0)	(4.5)	4	20	55

2030 check: ~\$90M revenue on ~9 average units ≈ \$10M per unit-year, above the \$2.4M OPEX break-even.

Scenario fan — 2030

CASE	UNITS	REVENUE	EBITDA
Conservative	6	\$45M	\$18M
Base	12	\$90M	\$55M
Upside	12+	\$130M	\$85M

8. Use of funds & milestones

USE	SHARE
Pathfinder + first full-scale hull	35–45%
CA firefighting + first metro modules	20–30%
Production tooling / learning-curve	15–20%
Ops bases, helium, insurance, crew	10–15%
Corporate / BD / compliance	5–10%

Gate heavy fleet CAPEX on endurance demo, modular swap demo, and first paid availability or mission contract.

9. Risks

- Airworthiness / airspace — pathfinder before fleet claims
- Early unit cost above \$13M volume target — learning-curve modeled explicitly
- Public contracting cycles — civil commercial utilization underwrites standby
- 17.8 : 1 ROI is a planning ten-year fleet figure, not a 2026–2030 guarantee

10. The ask

We are engaging venture capitalists and business owners to fund or co-develop:

1. Pathfinder validation and first full-scale hull
2. California firefighting concentration as the first revenue-bearing mission stack
3. Metro pre-positioning nodes that turn TITAN-X into infrastructure

Bottom line. Indefinite endurance. 20,000 lb modular payload. Near-zero persistent ops carbon. Pre-positioned regional coverage with a California firefighting emphasis. Dual-use economics on one hull family. 2026–2030 builds the twelve-unit base; utilization after that is where the summary ROI lives.

Prime Field Technologies LLC · TITAN-X Executive Business Plan · August 2026

Source lock: TITAN-X Consolidated Summary · Regional Deployment Concept. Planning estimates only.